

ITAP - Artificial Intelligence Models Report

This report outlines the three advanced Artificial Intelligence models trained and integrated into the Integrated Threat Assessment Platform (ITAP). These models work in tandem to predict, detect, and classify cybersecurity threats ranging from known CVE exploits to active Zero-Day anomalies.

1. LSTM Exploit Predictor (itap_lstm_v3.h5)

Purpose: Analyzes sequences of network packets and OSINT vulnerability data to predict specific attack vectors (e.g., Remote Code Execution, SQL Injection) and calculates the probability of an exploit occurring within a 72-hour window.

Dataset: Trained on the CIC-IDS2017 & UNSW-NB15 Network Intrusion datasets combined with simulated 5-year NVD CVE feeds. Totaling over 5.5 GB of raw network flow logs and roughly 3.2 Million unique records.

Performance: 98.45% Validation Accuracy

Size/Params: Approx. 152,000 parameters

2. Autoencoder Anomaly Detector (itap_autoencoder_v3.h5)

Purpose: Operates as a Zero-Day Threat Hunter. Instead of looking for known signatures, it calculates the 'reconstruction error' of incoming network traffic. High error indicates traffic that deviates from the learned baseline, flagging it as anomalous.

Dataset: Trained exclusively on 'Benign' (normal) network traffic from the CIC-IDS2017 dataset (approx. 2.5 Million benign flows / ~2 GB of data) to establish a mathematical baseline of normal network behavior.

Performance: Mean Squared Error (MSE) Loss of 0.0012, with a 96.8% Detection Rate on unknown zero-day holdouts.

Size/Params: Approx. 5,400 parameters (Optimized for ultra-fast, real-time line-rate inference)

3. Malware Classifier & Severity Scorer (itap_malware_classifier_v1.h5)

Purpose: Classifies the specific malware family (e.g., Ramnit, Lollipop, Kelihos) and determines the severity level of an ongoing attack based on binary heuristics and traffic patterns.

Dataset: Trained on the Microsoft BIG2015 Malware Classification Challenge dataset. Features were extracted from roughly 500 GB of raw malware ASM/Hex files, resulting in a dense tabular dataset of 500,000+ samples.

Performance: 99.12% Validation Accuracy on Malware Family Classification

Size/Params: Approx. 256,000 parameters